$$\delta F(x, y) = \frac{\partial F}{\partial x}\delta x + \frac{\partial F}{\partial y}\delta y$$

(derived from taylor series)

d = differential

δ = variation

$$\frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{q}_i} = \frac{\partial \mathcal{L}}{\partial q_i} \quad ; \quad \text{cdq } \mathcal{L} = \frac{1}{2}m\dot{q}_i^2 - V(q_i)$$

$$\frac{\partial \mathcal{L}}{\partial \dot{q}_i} = m\dot{q}_i = P_i$$

$$\Rightarrow \boxed{\dot{P}_i = \frac{\partial \mathcal{L}}{\partial q_i}}$$

$$\Rightarrow \boxed{P_i = \frac{\partial \mathcal{L}}{\partial \dot{q}_i}}$$

Symmetry ~~~ Conservation Laws

$$\delta q_i = f_i(q)\,\delta$$

$$\delta \mathcal{L} = \sum_i \frac{\partial \mathcal{L}}{\partial \dot{q}_i}\delta \dot{q}_i + \frac{\partial \mathcal{L}}{\partial q_i}\delta q_i$$

$$\delta \mathcal{L} = \sum_i P_i \delta \dot{q}_i + \dot{P}_i \delta q_i$$

$$\delta \mathcal{L} = \frac{d}{dt}\sum_i P_i \delta q_i$$

$$\delta \mathcal{L} = \frac{d}{dt}\sum_i P_i f_i(q)\,\delta$$

i.e. If symmetry $\longrightarrow \delta \mathcal{L} = 0$

$$\Downarrow$$

$$\frac{d}{dt}\sum_i P_i f_i(q)\,\delta = 0$$

$$\Downarrow$$

$$\sum_i P_i f_i(q) = \text{conserved}$$

Example :

i)

a) $\mathcal{L} = \frac{1}{2}m(\dot{q}_1^2 + \dot{q}_2^2) - V(q_1 - q_2)$

$$\left.\begin{aligned} \dot{P}_1 &= -V'(q_1 - q_2) \quad - (i) \\ \dot{P}_2 &= +V'(q_1 - q_2) \quad - (ii) \end{aligned}\right\} \because \frac{d}{dt}(P_i) = \frac{\partial \mathcal{L}}{\partial q_i}$$

Add (i) ; (ii) :

$$\frac{d}{dt}(P_1 + P_2) = 0$$

i.e. $(P_1 + P_2)$ is conserved.

b) generalized version of 'a'.

$$\mathcal{L} = \frac{1}{2}m(\dot{q}_1^2 + \dot{q}_2^2) - V(aq_1 - bq_2)$$

$$\dot{P}_1 = -aV'(aq_1 - bq_2) \longrightarrow b\dot{P}_1 = -ab\,V'(aq_1 - bq_2) \quad - (i)$$

$$\dot{P}_2 = +bV'(aq_1 - bq_2) \longrightarrow a\dot{P}_2 = +ab\,V'(aq_1 - bq_2) \quad - (ii)$$

Add (i) ; (ii) :

$$\frac{d}{dt}(bP_1 + aP_2) = 0$$

i.e. $(bP_1 + aP_2)$ is conserved

Made with Goodnotes